sustainability

MDPI

Systematic Review

Agroecology and Sustainable Agriculture: Conceptual Challenges and Opportunities—A Systematic Literature Review

Karla Terán-Samaniego [1], Jesús Martín Robles-Parra [2,*], Irasema Vargas-Arispuro [3], Miguel Ángel Martínez-Téllez [4,*], María Cristina Garza-Lagler [2], David Félix-Gurrlola [5], Mayra Lucía Maycotte-de la Peña [1], Julio César Tafolla-Arellano [6], Jorge Alan García-Figueroa [5] and Pablo César Espinoza-López [1]

1 Postdoctoral Stays, Department of Economics, Regional Development Coordination, Research Center for Food and Development (Centro de Investigación en Alimentación y Desarrollo, A.C./CIAD), Gustavo Enrique Astiazarán Rosas, No. 46, Hermosillo 83304, Sonora, Mexico; karla.teran@ciad.mx (K.T.-S.); mayra.maycotte@ciad.mx (M.L.M.-d.l.P.); pablo.espinoza@ciad.mx (P.C.E.-L.)
2 Department of Economics, Regional Development Coordination, Research Center for Food and Development (Centro de Investigación en Alimentación y Desarrollo, A.C./CIAD), Gustavo Enrique Astiazarán Rosas, No. 46, Hermosillo 83304, Sonora, Mexico; maria.garza@ciad.mx
3 Food Science Coordination, Research Center for Food and Development (Centro de Investigación en Alimentación y Desarrollo, A.C./CIAD), Gustavo Enrique Astiazarán Rosas, No. 46, Hermosillo 83304, Sonora, Mexico; iris@ciad.mx
4 Plant-Based Food Technology Coordination, Research Center for Food and Development (Centro de Investigación en Alimentación y Desarrollo, A.C./CIAD), Gustavo Enrique Astiazarán Rosas, No. 46, Hermosillo 83304, Sonora, Mexico
5 PhD Program in Regional Development, Department of Economics, Regional Development Coordination, Research Center for Food and Development (Centro de Investigación en Alimentación y Desarrollo, A.C./CIAD), Gustavo Enrique Astiazarán Rosas, No. 46, Hermosillo 83304, Sonora, Mexico; dfelix423@estudiantes.ciad.mx (D.F.-G.); jgarcia423@estudiantes.ciad.mx (J.A.G.-F.)
6 Laboratory of Biotechnology and Molecular Biology, Basic Sciences Department, Antonio Narro Agrarian Autonomous University, Calzada Antonio Narro, Buenavista, No. 1923, Saltillo 25315, Coahuila, Mexico; jtafare@uaaan.edu.mx
* Correspondence: jrobles@ciad.mx (J.M.R.-P.); norawa@ciad.mx (M.Á.M.-T.); Tel.: +52-6622892400 (ext. 373) (J.M.R.-P.); +52-6622892400 (ext. 409) (M.Á.M.-T.)

check for updates

Academic Editors: James W. Muthomi, Alex M. Fulano and Nancy Karimi Njeru

Received: 6 December 2024
Revised: 21 January 2025
Accepted: 30 January 2025
Published: 20 February 2025

Citation: Terán-Samaniego, K.; Robles-Parra, J.M.; Vargas-Arispuro, I.; Martínez-Téllez, M.Á.; Garza-Lagler, M.C.; Félix-Gurrlola, D.; Maycotte-de la Peña, M.L.; Tafolla-Arellano, J.C.; García-Figueroa, J.A.; Espinoza-López, P.C. Agroecology and Sustainable Agriculture: Conceptual Challenges and Opportunities—A Systematic Literature Review. *Sustainability* **2025**, *17*, 1805. https://doi.org/10.3390/su17051805

Copyright: © 2025 by the authors. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license (https://creativecommons.org/licenses/by/4.0/).

Abstract: The transition of agriculture towards sustainability faces significant obstacles, such as increased demand for food, food insecurity, climate variability, biodiversity loss, and food waste, among others. Moreover, agricultural activities must address ethical practices within sustainable development. The literature frequently mentions two approaches to meet these challenges: agroecology and sustainable agriculture. This study aims to delineate the conceptual boundaries of agroecology and sustainable agriculture while elucidating their interconnection. It seeks to clarify the scope and limitations inherent in these agricultural practices, which is critical given the centrality of sustainable agriculture in agrarian studies. By establishing these boundaries, the research outlines the methodologies to identify the relevant variables and indicators required for effective stakeholder engagement within agricultural systems. A systematic literature review was conducted using the PRISMA method. The databases searched were Science Direct, Scopus, Nature, and Google Scholar. The inclusion criteria were (i) written in English or Spanish, (ii) published in a peer-reviewed academic journal, and (iii) related to the conceptualization of agroecology and sustainable agriculture. Publications were selected following the method's identification, screening, eligibility, and inclusion guidelines. The main distinctions between the concepts are the scale and the scope. Agroecology reveals limitations to achieving impacts on food security and agriculture's challenges in a large-scale system better suited for rural communities and small farms seeking locally adapted solutions. On the other hand, the literature reveals that it is crucial to adopt a multi-scale systems approach to meet the

growing food demands of the global population, for which sustainable agriculture may be more effective.

Keywords: agroecology; sustainable agriculture; sustainable development; agricultural systems

1. Introduction

Agriculture encounters environmental, economic, and social challenges affecting small- and large-scale production. It is essential to comprehend the principles of agroecology and sustainable agriculture and explore how implementing these concepts can help ensure food security [1]. Contributions to agroecology encompass various definitions, from approaches based on a single discipline to complex concepts that integrate interdisciplinary ideas [2–30]. This includes links between fields of knowledge (interdisciplinarity), different ideas within the exact science (intradisciplinarity), and the association of scientific concepts with traditions, culture, and governance to understand or address social issues (transdisciplinarity), including the significant question of food security [7,8,11,12]. Sustainable agriculture is defined by research as an agricultural activity based on a production system capable of maintaining productivity in the long term [5,12,14,15,23]. The objectives are to provide food for people at reasonable prices and to be profitable enough to compete with conventional agriculture without compromising the potential of natural resources. This contributes to food security. In this context, it is essential to note a lack of in-depth analysis of agroecology and sustainable agriculture concepts in the existing literature. The contributions made so far have been quite general, with authors seldom taking a well-supported stance [6,12]. Therefore, it is essential to explore further how these concepts are defined, the disciplines that shape them, their goals, and the individuals and groups involved, among other aspects. This work involves systematically searching for information to establish the conceptual foundation of the terms mentioned, defining what each term entails and what it does not. The researchers aim to clarify the scope and limitations of agricultural practices, identify the elements, characteristics, and agents involved in each concept, and examine their contributions to food security. The concept of agroecology does not tacitly consider the scale of production and the market perspective, which are essential in the area of sovereignty, food security, and agricultural sustainability [16,24–28], and it is precisely these conceptual interactions that are the main contribution of this article, which is equally consistent with the referential gap. This conceptual framework assumes a methodological approach combining qualitative and quantitative perspectives, adopting a mixed-methods strategy.

2. Agroecology or Sustainable Agriculture?

The idea of agroecology originated from combining two fields, agronomy ("agro") and ecology, to tackle the challenges stemming from agricultural activity. However, as farming systems are complex and their issues need urgent resolution, additional diverse fields of study have been incorporated. This indicates a shift in the concept, which has become constrained by its name. In light of this, the ideas of agroecology and sustainable agriculture will be examined based on the contributions of various authors.

2.1. The Evolution of the Concept of Agroecology

Various experts have documented the description and theoretical development of agroecology. Some researchers trace its origins back to the 1920s, when it was defined as the application of agronomy and ecology approaches to the study of commercial crop cultivation [2–4] (see Figure 1). During the following three decades, contributions to

agroecology showed minimal variation. Scientific contributions focused on applying and understanding natural processes to enhance crop productivity and manage pests in specific production areas [5].

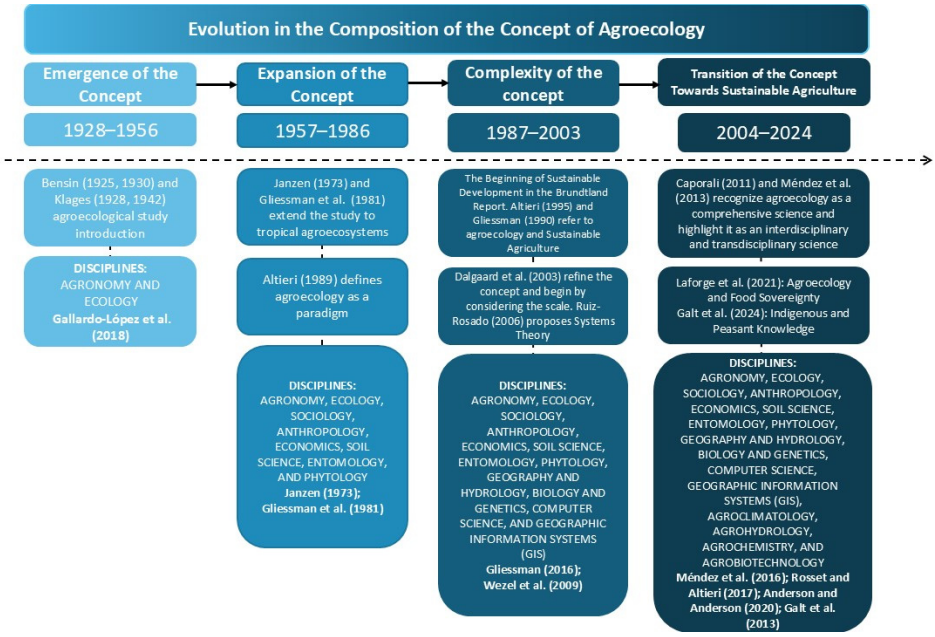

Figure 1. Evolution in the composition of the concept of agroecology. Based in [2–4,7,8,10,12,14–16,18,19,21–24,26–29,31–33].

Before the 1960s, there was a significant change in how people viewed the relationship between humans and the natural world. This led to a greater understanding of the importance of using methods of food production that work in harmony with the environment. At the same time, agroecology mostly aligns with social movements, as opposed to historical and scientific events like the Green Revolution [6].

Between 1957 and 1986, Janzen [7] and Gliessman et al. [8] researched tropical agricultural systems. They emphasized the importance of incorporating disciplines such as sociology, anthropology, economics, soil science, entomology, and plant science into agroecology. They highlighted the significance of sociology and anthropology in maintaining a balance in production and its interaction with ecological and socioeconomic factors, as well as the incorporation of soil science, which is crucial for soil health and crop diversity, entomology for utilizing insects in organic biomass content, and plant science for ensuring plant health and well-being (see Figure 1).

On the other hand, Levidow et al. [9] first noted that agroecology began as a field that rigorously examines ecological relationships within agricultural systems but has grown in complexity and focus in recent decades. In this context, Ruiz-Rosado [10] agrees and argues that the emerging conceptual and practical agroecological literature since the 1970s provides initial evidence of a shift towards systems theory.

This theoretical framework emphasizes the importance of viewing an agroecosystem holistically. It recognizes that an agroecosystem is more than just the sum of its parts, and it aims to establish a framework that shapes agroecological thinking by drawing on theoretical contributions from various disciplines. This systems approach allows for exploring agricultural aspects beyond ecological dynamics [11].

Caporali [12] and Fernández [13] emphasize the importance of considering agroecology as a system in their research, just as Levidow et al. [9] and Ruiz-Rosado affirmed [10]. This is because it involves human actions, leading to various scenarios and requiring input from multiple disciplines. They also highlight the need to integrate an ethical dimension

that can address problems stemming from human influence over the environment, resource exploitation, and environmental degradation in the modern era.

Sustainable development was introduced in the Brundtland Report between 1987 and 2003. This paradigm aimed to integrate financial, social, and environmental elements to enhance social well-being and reduce ecological impact, influencing how each science is viewed, including agronomy [14]. Altieri [14] and Gliessman [15] refer to the concepts of agroecology and sustainable agriculture, but they do not provide definitive specifications regarding their respective definitions. To address this necessity, Dalgaard et al. [16] and Francis et al. [17] attempted to clarify the concept of agroecology in their research. At the same time, Ruiz-Rosado [10] proposes a systems approach to address sustainable agriculture in agricultural systems (see Figure 1).

Figure 1 shows the contribution of Gliessman [18] and Wezel et al. [19], who have conducted a thorough analysis of the concept of agroecology, emphasizing the necessity for the explicit inclusion of disciplines such as the following:

i. Geography and Hydrology: Although not detailed, these are implicit in the overall redesign of agroecosystems and managing resources such as water.
ii. Biology and Genetics: These aspects of improved seeds and nitrogen-fixing crops are mentioned, indicating that they are practices from the biological sciences aimed at sustainable agriculture.
iii. Computer Science and Geographic Information Systems (GIS) are essential for optimizing input use and managing agricultural practices more efficiently. This modern approach heavily relies on data and technology to enhance agricultural productivity.

The conceptual framework has been continuously developed into the early 21st century. This is a crucial era during which publications such as those by Bezner et al. [20] delve into agroecology's complex nature in food production. In this regard, Méndez et al. [21] emphasize the importance of building the research topic from diverse interdisciplinary perspectives and highlight the indispensable role of systemic thinking in theoretical advancements, which accords with Ruiz-Rosado [10].

In the past twenty years, agroecology has emerged as an academic field that promotes sustainable food systems through interdisciplinary support, incorporating alternative ideas such as indigenous and peasant knowledge [22]. These ideas are essential for aligning food production with the conservation of natural resources and social justice [16]. They should be grounded in scientific and philosophical principles to foster agricultural sustainability [23].

On the other hand, Dalgaard et al. [16] suggest that scale must be an essential aspect of agroecology, which is seen as a transdisciplinary field within a food system aiming for sustainability. The scholars emphasize that the development of agroecology has advanced from focusing on individual plots or fields (from the 1930s to the 1960s) to a broader agroecosystem level (from the 1970s to the 2000s).

Laforge et al. [24] acknowledge that agroecology should be seen as a framework for food sovereignty in today's context. It addresses the communities' right to access nutritious and culturally appropriate food cultivated using environmentally sustainable practices. This concords with López-García et al. [25]; food sovereignty is crucial for research in agroecology, but it is also food security that makes it necessary to focus on addressing challenges related to scalability and transdisciplinarity. Collaborative efforts are essential for promoting sustainable food production systems and establishing agroecology as a field capable of significantly contributing to global sustainability.

In this context, it is pertinent to recognize the research conducted by Anderson and Anderson and Anderson [26], Galt et al. [27], Méndez et al. [28], and Rosset and Altieri [29]. They discuss the adaptation of agricultural systems to local climatic and hydric conditions and the application of agrochemical and biotechnological principles in agroecology to

enhance the sustainability of farming systems, considering specific needs. This underscores the necessity of incorporating disciplines such as agroclimatology, agrohydrology, agrochemistry, and agrobiotechnology. These references illustrate how the document integrates a wide range of disciplines in its approach to agroecology, highlighting its transdisciplinarity and participatory nature (see Figure 1).

In this situation, Rossi [30] emphasizes that agroecology extends beyond theoretical boundaries as its approach aims to be more holistic. This contrasts with the foundational concepts of the terms "agro" (from agronomy) and "ecology", which have become limited in addressing agricultural issues due to the need to incorporate new disciplines.

The evolution of agroecology can be traced through a series of critical developments. In the 1930s, the focus was on applying ecological principles to agriculture. Between 1957 and 1986, agroecology was considered a paradigm, with a particular emphasis on tropical agroecosystems. From 1987 to 2003, the concept of sustainable development was introduced, leading to a shift towards systems theory. Since 2004, agroecology has incorporated essential components such as scaling, interdisciplinarity, transdisciplinarity, food sovereignty, and food security [2–33] (see Figure 1).

The characteristics explained in this section make it clear that agroecology integrates various disciplines and approaches depending on the issues to be addressed [10]. However, the agronomic or ecological perspectives have been insufficient in addressing the need to consider a perspective that includes each context's social, economic, and political dimensions [22].

In other words, the discussed concept must account for humans' relationship with organisms, ecological systems, and the environment in which agricultural activities occur. In this scenario, the individual makes decisions and transforms agroecosystems based on information obtained from other farmers, agricultural producers, media, educational and research institutions, or service, financial, or political institutions [9,10,12,13].

For this reason, the combination of the two components "agro" and "ecology", corresponding to the sciences of "agronomy" and "ecology", respectively, has acquired a theoretical breadth in which it is evident that the term "agroecology", based on its conceptual foundation, has become limited [10]. Hence, the need arises to rename agroecology to sustainable agriculture (Figure 2).

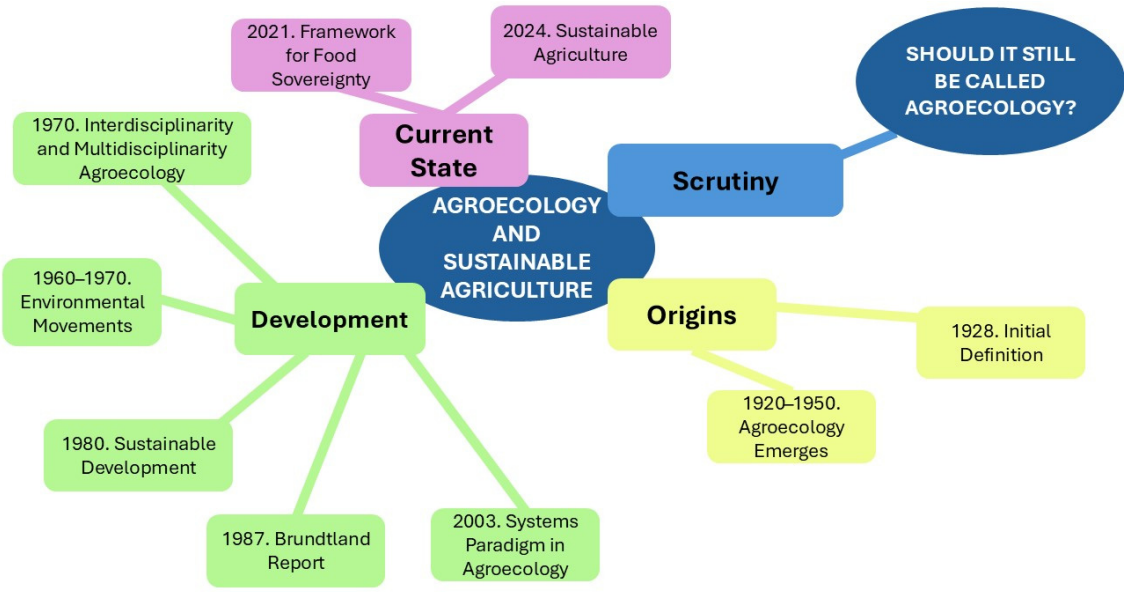

Figure 2. Framework for whether to continue calling it agroecology when the concept has evolved into sustainable agriculture.

2.2. Sustainable Agriculture

The evolution of sustainable agricultural practices has been shaped by various factors stemming from the Green Revolution, which unfolded in the latter half of the 20th century (1940–1950). This revolution brought about environmental and social impacts, generating a growing interest in agricultural and food systems [34,35].

In 1962, Rachel Carson published her book "Silent Spring", which played a pivotal role in the modern environmental movement. The book brought attention to the detrimental effects of pesticides on the environment and sparked widespread scrutiny of intensive agricultural practices. Eight years later, global environmental awareness increased significantly following "The First Earth Day". This event addressed various environmental crises and emphasized the necessity for more sustainable practices, particularly in agriculture [36].

The concept of sustainable development was defined by the World Commission on Environment and Development in the 1987 Brundtland Report [37]. In 1990, the term gained formal academic and political recognition, leading to increased research and programs focused on sustainable agriculture [38]. Additionally, during the 1992 Earth Summit in Rio, Agenda 21 was established, featuring dedicated chapters on sustainable agriculture [39].

During the 2000s, growing concerns from consumers and non-governmental organizations about the environmental and health effects of harmful agricultural production and consumption resulted in the creation of standards and certifications like GlobalG.A.P. (formerly EUROPG.A.P.), Fair Trade, and Rainforest Alliance. These initiatives aim to encourage sustainable agricultural practices [40].

In the 2010s and 2020s, technology became integrated into sustainability initiatives. Advances in agricultural technology, including precision agriculture and agro-technology, showcased the necessary support for sustainable practices. Moreover, during this period, organizations like the Food and Agriculture Organization (FAO) noted an uptick in acknowledging agroecology as an approach to achieving agricultural sustainability [41]. With the global climate crisis and food insecurity becoming critical issues, sustainable agriculture is increasingly considered a vital solution.

This timeline provides an overview of the development and evolution of sustainable agriculture, showing how it has been shaped by environmental, political, and technological changes over time (Figure 3).

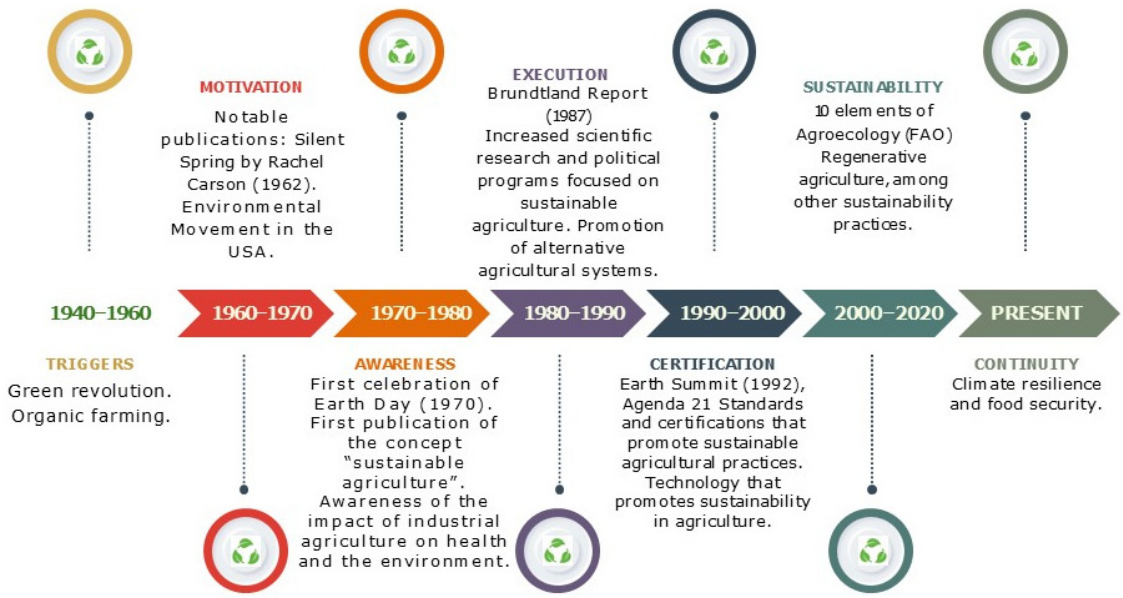

Figure 3. Sustainable agriculture timeline.

The concept of sustainable agriculture is multifaceted, involving the interplay of technological advances, policies, and social dimensions. Its development has been influenced by various factors, including increased environmental awareness, shifts in global food demand, technological innovations, the incorporation of traditional knowledge, and changes in political frameworks [35].

Sustainable agriculture can be defined as an agricultural production system designed to conserve resources, prioritize environmental responsibility, and remain economically viable. Its goal is to meet the food needs of present and future generations, preserve natural resources, utilize agricultural and energy resources efficiently, adapt to natural biological cycles, and foster rural economic development while enhancing the quality of life for farmers [42].

This concept can be advanced through various means, including developing appropriate technologies, agrarian reform to improve access to productive resources, crop rotation, organic fertilizers, agroforestry (integrating agriculture and livestock), and Integrated Pest Management. This approach is endorsed by international organizations, non-governmental organizations, farmer networks, and research centers that seek to promote agricultural practices that are more mindful of the environment and local communities [43].

The term has emerged from the necessity to establish alternative agricultural methods to better align with environmentally friendly production practices. The primary objective of sustainable agriculture is to meet human dietary needs while enhancing environmental quality, conserving natural resources, effectively utilizing agricultural resources and non-renewable energy sources, adapting to natural biological cycles, and fostering rural economic growth and the well-being of farmers [44].

The definition of sustainable agriculture, as articulated by the authors of this paper, refers to an agricultural approach that embraces the principles of sustainable development to address the food needs of present and future generations. This is achieved through agricultural production that safeguards the environment, responsibly utilizes agricultural and energy resources, aligns with the biological cycles of ecosystems, and fosters the development, economy, and well-being of farmers and communities, thereby contributing to food security [42–44].

For this research, agroecology is an interdisciplinary, intradisciplinary, transdisciplinary, and systemic approach supported by sustainable food production systems to meet scaling needs. This comprehensive approach to agroecology aligns with the concepts of food security [7–9,11,12,16,21,22,25–29].

What do agroecology and sustainable agriculture encompass or exclude, and what factors do they consider or disregard? Where are the deficiencies and misunderstandings in using these crucial concepts for agricultural practices?

Mockshell and Kamanda [45] discuss in their research that agroecology and sustainable agriculture share the common goal of promoting food security through the optimal and sustainable use of resources. However, they point out that the stakeholders involved in these approaches have different perspectives on various aspects, such as the concept of agriculture, the role of genetic engineering, the scale of operation, land use, soil health, and their views on the socio-economic and ecological dimensions of sustainable development.

To address this question, the authors conducted a systematic literature review focusing on the economic, social, and environmental aspects of sustainable development related to agriculture. They made a clear distinction between what is commonly included in both concepts. The areas and indicators considered are as follows: general (concept, main actors, vision, science, and opposing viewpoints); the environmental dimension (land use, space allocation, landscape, and agricultural practices); the social dimension (knowledge

generation, farmers, and support for livelihoods); and the economic dimension (efficiency, seed system, and input use). You can refer to Table 1 for more details [45].

Table 1. Differences between the concepts of agroecology and sustainable agriculture. Adapted from Mockshell and Kamanda [45].

Areas	Indicators	Agroecology	Sustainable Agriculture
General	Concept	It aims to increase agricultural production and capitalize on ecological processes that conserve natural resources in agroecosystems. It seeks to use a holistic approach to rural development that includes all environmental and human elements, as well as employ a set of practices to mimic nature. It attempts to present agriculture as an 'ecology of disciplines'.	It aims to increase agricultural productivity while simultaneously protecting natural capital. It focuses on resource intensification and efficiency in their use. It seeks to meet the needs of the present generation without compromising the ability of future generations to meet their own needs.
	Key Actors	Non-Governmental Organizations (NGOs), civil society, researchers, and academics.	Governments, multinational agroindustries in the private sector (agrochemicals, fertilizers, seeds), researchers, academics, and international development institutions.
	Vision	Food security, development in favor of the poor, and environmental sustainability. It seeks to be a sustainable alternative compared to the hegemonic style of conventional and agroindustrial agriculture.	Food and nutritional security, poverty reduction, and environmental, social, and economic sustainability. It aims to be an alternative to conventional intensification or industrial agriculture.
	Science	GMOs are not acceptable.	GMOs may be tolerated.
	Opposition Stances	Considered by its opponents as 'anti-science' and 'a do-nothing approach'.	Considered by its opponents as a 'model of conventional and industrial agriculture', that is, 'the usual'.
Environ-Mental	Land Use	Land sharing (less land is reserved for conservation, and less intensive production techniques are used to maintain biodiversity).	Land sparing (reserving land for intensive production while setting aside part of the land for biodiversity).
	Spatial Arrangement	Mixed agriculture and multifunctional crops, mixed crop and livestock systems.	Monoculture.
	Landscape	Build resilient agroecosystems through ecosystem services.	Minimize environmental damage through intensification rather than area expansion.
	Agricultural Practices	Biological interactions in diversified agricultural systems to enhance productivity.	Combination of improved varieties and agronomy with good agricultural practices.
Social	Knowledge Generation	Generally based on local knowledge.	Generally based on local and expert knowledge.
	Farmers	Small-scale farmers.	Large-scale farmers.
	Livelihood support	Rural households of smallholder farmers.	Companies and collaborators of large-scale producers.
Economic	Efficiency	Land use efficiency (yield). Yield gap/yield potential gap.	Agricultural productivity and alternatives consistent with sustainable development.
	Seed system	Local seeds (homegrown seeds or seed exchange system).	External seeds (seed industry).
	Input use	Low use of external inputs (low cost).	High use of external inputs (high cost).

Mockshell and Kamanda [45] indicate that the elements of agroecology and sustainable agriculture work towards achieving the sustainability of the entire agricultural system.

3. The Transition to Agroecology and the Evolution Towards Sustainable Agriculture

The shift to agroecology and the advancement of sustainable agriculture are encountering significant hurdles, such as increasing food demand, food insecurity, nutritional deficiencies, climate variability, environmental resource depletion, biodiversity loss, and food waste. Achieving sustainable agriculture necessitates prompt global interventions and evidence-based strategies to yield tangible results [46].

The Economic Commission for Latin America and the Caribbean (CEPAL) underscores in a document the importance of employing tools, approaches, and a specialized advisory, as well as fostering collaboration among producers, scientists, and experts, to implement agroecology as a model for agricultural production and advance towards sustainable agriculture [47]. It is an interdisciplinary field that aims to enhance sustainability across various dimensions. It involves an approach to agricultural production and the agri-food system that prioritizes participatory development to achieve agricultural sustainability and a more democratic decision-making process in food production [48].

An examination of various documents reveals that integrating fundamental principles of agroecology and sustainable agriculture on a global scale, particularly in Latin American countries, remains inadequate. This shortfall can be attributed primarily to several factors: (1) the prioritization of practical and timely resolutions over theoretical exploration; (2) the limited availability of informational resources abroad; (3) the insufficiency of financial resources for publishing and disseminating printed materials, as well as for evaluating digital content; and (4) the prevailing adherence to the productivity paradigm of conventional agriculture among critical actors in academic, commercial, and political spheres [49].

In line with the Sustainable Development Goals (SDGs), ensuring sustainability in agriculture is recognized as a top priority in tackling the critical challenge of attaining food security and promoting healthy nutrition for all. This necessitates addressing environmental and social progress as a primary focus. Achieving sustainability in agricultural systems requires a long-term view and comprehensive approaches, which are increasingly acknowledged for their capacity to drive transformative change in agriculture [50].

The principles of agroecology cover normative and relational aspects. They can be applied to different scales (fields, farms, landscapes, or entire food systems) and various dimensions of food systems, including production and governance. Today, agroecology is linked with a set of criteria for the agricultural and ecological management of agri-food systems, as well as broader socio-economic, cultural, and political elements [25,45,49]. Gallardo-López et al. [4] agree and point out that agroecology has progressed from focusing on individual plots or fields in the 1930s–1960s to agroecosystems from the 1970s–2000s.

In summary, the authors of this article believe that agroecology is an interdisciplinary, intradisciplinary, transdisciplinary, and systemic approach that assists agricultural activities in transitioning towards sustainable agriculture, influenced by the three domains of sustainable development [7–9,11,12,16,21,22,25–29].

Agricultural practices must encompass ethical considerations within sustainable development, large-scale production, and food security. It is also essential to meet the present generation's needs without compromising the ability of future generations to meet their own. Achieving these goals requires the active involvement of key stakeholders across different sectors and regions, including governments, agribusinesses, researchers, companies, and non-governmental organizations [44,45].

Agroecology lacks a systems perspective that considers small-scale and large-scale production units, making it challenging to contribute fully to sustainable and resilient agricultural systems. Agroecology emphasizes sustainable agriculture by utilizing ecosystem services and promoting biodiversity [25,45,49]. On the other hand, sustainable agriculture

relies on various practices to ensure the long-term environmental, social, and economic functioning of complex agricultural systems [42–44].

Sustainable agriculture involves the preservation of biodiversity and ecological processes, promoting human health and increasing productivity while minimizing negative impacts on the environment and economic benefits [42]. In contrast, agroecology primarily focuses on the environmental aspects of sustainable development [9]. Sustainability in agriculture encompasses broader practices that extend beyond the principles of agroecology.

Thus, what is needed globally to integrate the principles of sustainable development into agricultural practices, advance toward sustainability, and contribute to food security is applying the concept of sustainable agriculture to complex agrarian systems across different types of production units and at various scales [16,24–28].

4. Methodology

This systematic literature review followed the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines [51–53], was carried out in October 2024, and used the ScienceDirect, Scopus, Nature, and Google Scholar databases. These databases serve as dependable and specialized sources of information for technical and scientific research, featuring an extensive compilation of scientific journals and articles. They encompass a multidisciplinary approach, addressing a diverse range of academic disciplines. The PRISMA checklist (Table S1) is included in the Supplementary Material. Additionally, this review was executed in four phases.

In all phases of the study, specific exclusion criteria were established to ensure the relevance and rigor of the selected literature. Excluded materials encompassed book chapters, manuscripts, opinion pieces, viewpoints, anecdotes, letters, theses, dissertations, conference papers, editorials, or articles unrelated to the research objective, not written in English or Spanish, and non-full access.

Conversely, the inclusion criteria mandated those eligible articles be published in specialized, peer-reviewed journals and written in English or Spanish—proficiently spoken by the authors, include keywords or a combination of them, as well as define, analyze, or discuss the concepts or definitions of agroecology or sustainable agriculture, their elements, characteristics, agents, limitations, and contributions. This approach was designed to enhance the quality and applicability of the research findings.

Phase one focused on the keywords: "agroecology concept"; the second phase included "sustainable agriculture concept". For the third phase, the keywords were "agroecology definition", and for the fourth phase, the keywords used were "sustainable agriculture definition".

The first search yielded 18 articles in ScienceDirect, 4 in Scopus, 92 in Nature, and 146 in Google Scholar. The second search identified 30 articles in ScienceDirect, 18 in Scopus, 2310 in Nature, and 389 in Google Scholar. The third search produced 2 articles in ScienceDirect, 0 in Scopus, 50 in Nature, and 29 in Google Scholar. The fourth search resulted in 17 articles in ScienceDirect, 2 in Scopus, 1680 in Nature, and 318 in Google Scholar (see Figure 4).

Four authors dedicated themselves to research information. Exclusion criteria were applied to the searches. Data items from each article, such as title, author, and year, were captured in a database. Two different authors independently reproduced the searches to confirm the information obtained and avoid biases in the identification process. All the results found were combined eliminating duplicates. As a result, 123 articles were selected upon applying the exclusion criteria. Titles of these articles were reviewed, discarding those that did not include the keywords or combination of them. The abstracts of the remaining 52 articles were analyzed, eliminating those that were not relevant to the objective of the

review, and then the introductions of 18 articles were subsequently studied, ensuring that the topic of interest was directly addressed and provided an appropriate contextual framework. A total of 16 relevant studies were selected for full reading. From these readings, seven articles were eliminated. Only nine articles met the inclusion criteria and were considered for the review.

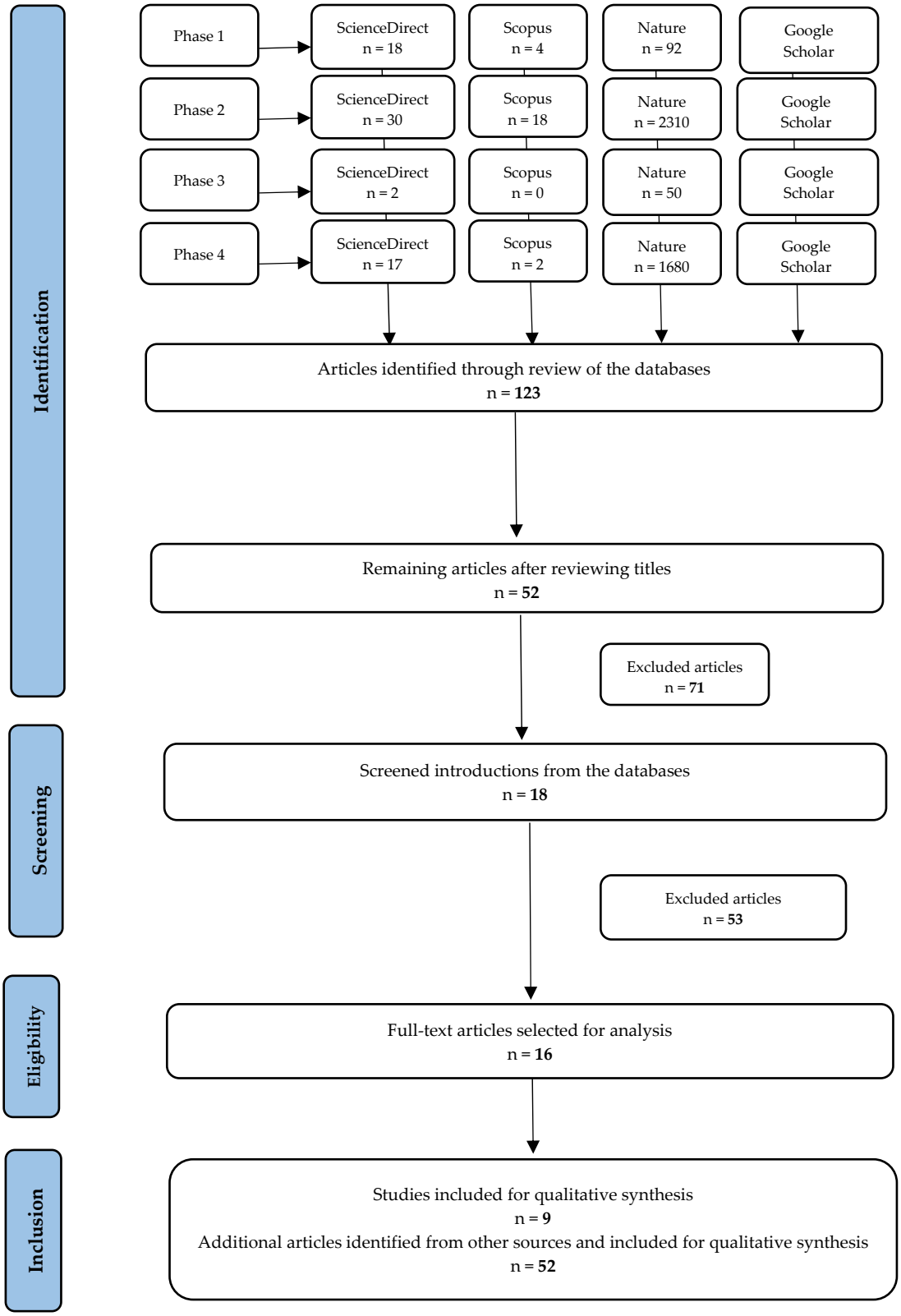

Figure 4. Flowchart of the systematic information search.

In Figure 4, the flow diagram used for the identification, screening, and inclusion of studies is presented. Excluded studies did not meet the inclusion criteria, they were not related to the subject of the objective, with topics such as the arrangement of data on precision agricultural technologies; the analysis of sustainability metrics; the comparative evaluation of environmental impact; the sustainability of systems through the Sustainable Development Goals; and the creation of composite indices of agricultural sustainability, among others.

This review aims to determine what the concepts of agroecology and sustainable agriculture entail and do not entail by identifying the contributions of various authors who have included different disciplines and characteristics in their definitions. This is because concepts are formed through the construction of knowledge and are enriched by discoveries. The review also illustrates the relationship between agroecology and sustainable agriculture, the actors involved, their perspectives, and their contributions to the three domains of sustainable development. The findings derived from the literature review are presented in Section 5.

This conceptual framework necessitates a methodological approach that integrates qualitative and quantitative perspectives, commonly called a mixed-methods approach. Such integration is essential, given that the foundational concepts require diverse variables and indicators that align with complexity and structuralism paradigms. Consequently, using an appropriate technique to synthesize these perspectives is imperative. This research proposes incorporating the Max QDA program, drawing from empirical experience, to facilitate this mixed-methods approach effectively.

5. Results

The authors independently analyzed the nine studies, and the information is presented in Table 2. This table includes the objectives and findings regarding the definitions of agroecology and sustainable agriculture and the perspectives, actors, and characteristics. The implications of these findings for the future of agriculture and sustainability are also discussed.

Regarding the definition of agroecology, Aguilera et al. [54] understand it as a valuable approach for addressing agro-environmental issues due to its holistic perspective. They emphasize that this comprehensive view considers the agricultural dimensions and the food system, including environmental and socioeconomic aspects. This perspective is essential for dealing with challenges across different dimensions of agriculture.

From the above, it is essential to highlight that the authors note that agroecology is a dynamic concept that has integrated socioeconomic, cultural, and political considerations, enriching its conceptualization to accommodate diverse cultural regions and sociopolitical contexts. They argue that agroecology is constantly evolving, encompassing a wide range of locally adapted practices contributing to the sustainability of agricultural systems [54].

From a productivity-focused perspective, Ewert et al. [55] advocate agroecology as a feasible alternative for managing and sustaining agricultural production yields. They argue that its foundation is rooted in natural ecosystems. The authors acknowledge that adopting agroecological practices presents financial uncertainty for farmers, which may hinder their large-scale implementation. This economic uncertainty could stem from the initial investment required for transitioning to agroecological practices and the potential fluctuations in yields during the transition period. They emphasize that its potential lies in optimizing internal environmental processes to enhance the functionality and adaptability of the agricultural system.

Muhie [56] points out that sustainable agriculture can be described using ecocentric and technocentric approaches. He argues that sustainable agriculture is connected to

food security, which involves sufficient food production, access to food and the ability to purchase it, nutritional adequacy, safety, and economic stability, which concord with a productivity perspective. The author explains that sustainable agriculture has the potential to directly contribute to achieving several Sustainable Development Goals (SDGs). These include goals related to poverty, hunger, inequality, responsible consumption and production, climate change, ecosystems, local and national development, and the environment.

Trigo, Marta-Costa, and Fragoso [57] thoroughly examine the conceptual and practical issues of using sustainable development as a standard or ethical principle to drive agricultural change. They highlight that this can be achieved through a system ownership perspective, a philosophical approach, or a measurable criterion. Even so, there is ongoing debate surrounding the low productivity and efficiency of agricultural systems that are considered to be sustainable. The critical question is whether it will be possible to maintain and enhance food production for future generations. The approaches used to achieve sustainability in agriculture are constantly being questioned due to the uncertainty surrounding ensuring food security. These last authors agree with Muhie [56], who establishes that sustainable agriculture primarily considers ecology and technology as essential components within its strategies and practices. They also mention that this is generally indicated by ideological interpretations of sustainability that tend to deviate from what could improve agricultural system performance.

However, Ogemah [58] still points out some doubts about the concept and reviews various definitions of sustainable agriculture. To organize ideas and contributions, he first establishes what sustainable agriculture is but not before defining what it is. Sustainable agriculture differs from organic agriculture, as the latter is just one part of sustainable agriculture. It is not a particular method of agricultural production, and it does not refer to a specific set of practices. Sustainable agriculture is not necessarily labor-intensive and does not rely solely on low-cost inputs or low-technology approaches. It does not mean farming without using fertilizers and synthetic chemicals, nor does it refer to agriculture with minimal external inputs or conservation agriculture. The author refers to the National Academy of Sciences' definition of sustainable agriculture.

According to Ogemah [58], the definition of sustainable agriculture does not rule out the use of technologies or practices for ideological reasons. Instead, it focuses on improving productivity to ensure food security without significantly impacting other Sustainable Development Goals. He argues that agriculture is sustainable when it is ecologically sound, economically viable, socially just, culturally appropriate, and based on a comprehensive scientific approach. It should also follow these principles: improvement in quality of life, a paradigm based on a systemic perspective, provides cost-effectiveness, must be managed, emphasizes science and experience, has a multidisciplinary foundation, and is an evolving process.

On the other hand, Hrabanski and Le Coq [59] have significantly contributed to agroecology. They note that while the concept originated in the 1920s, the foundational work was carried out by Altieri in 1983, who defined agroecology as a scientific discipline bridging agronomy and ecology. They also observe that perceptions of agroecology can vary widely from one country to another; for some, it is seen as a scientific field, for others, as a practical application, and, to a lesser extent, as a social movement. The authors suggest that the term "agroecology" refers to the necessity of integrating agronomy with ecology. They also argue that the understanding of this concept is rooted in history. Still, in recent decades, its advancement has been propelled by including other disciplines and various components of agri-food systems.

Likewise, Gil et al. [60] discuss agroecology and stress its significance in areas where industrial agriculture's adverse environmental and social effects are especially severe. They

highlight that the excessive exploitation of natural resources has caused environmental degradation, including salinization and soil fertility loss, ultimately leading to land desertification and complicating food security. The authors recommend transitioning to agroecology, creating more resilient and energy-efficient landscapes to strengthen local capacity for climate change mitigation and adaptation. It also involves increasing soil organic matter regeneration, ensuring freshwater supply, and enhancing biodiversity.

The systematic literature review by Mockshell and Kamanda [45] revealed differences in concepts, perspectives, actors, and characteristics of agroecology and sustainable agriculture. However, both aim to promote food security, a crucial research finding. These experts contend that sustainable agriculture takes a holistic approach, focusing on individual agricultural practices to tackle conventional agriculture's ecological, economic, and social issues. It aims to meet the increasing global food demand with minimal environmental impact. As for agroecology, they define it simply as applying ecological concepts and principles in the design and management of sustainable food systems. In addition, it discusses how agroecology supporters believe it can bring about a fundamental shift to boost farmers' production in developing countries. These regions often have limited access to external inputs, and a significant portion of the population suffers from food insecurity. Food self-sufficiency is central to this discussion, as it underscores the importance of local production in meeting internal food consumption needs.

Given this, it is also relevant to consider the findings of Amrit, Minakshi, and Raghavendra [61], who maintain that sustainable agriculture integrates traditional and modern scientific knowledge. This integration includes continuous coordination among various actors, disciplines, and elements to enhance the environmental, social, and economic aspects of the agri-food system. These are considered the key pillars of sustainable agriculture. These scholars emphasize that maintaining soil quality and health is crucial for long-term crop productivity. Sustainable agriculture is the only viable approach to meet the food demands of the growing global population and achieve food security.

The methodology proposed was applied to the search, and no evidence of express conceptual discussion of the terms studied was encountered. It was found that frontier studies on agroecology are focused on agroecological practices [62,63], transition [64,65], and study cases [66]. According to sustainable agriculture, the issues are mainly linked to novel practices [67,68]; trends, opportunities, and progress [69,70]; the analysis of the main challenges associated with the construction of policies for the reduction in inputs [71,72]; and study cases [73].

Table 2. Studies included in the qualitative synthesis, derived from the search using the following queries: phase 1 ("agroecology concept"), phase 2 ("sustainable agriculture concept"), phase 3 ("agroecology definition"), and phase 4 ("sustainable agriculture definition").

Reference	Objective	Definitions, Aspects, and Characteristics of the Concepts of Agroecology and Sustainable Agriculture	Results
Hrabanski and Le Coq [59]	To analyze three concepts that emerged from establishing the climate agenda: "climate-smart agriculture", "agroecology", and "nature-based solutions".	Agroecology definition.	They demonstrated that these concepts specifically frame the issues of agricultural adaptation and mitigation related to climate change.
Aguilera et al. [54]	To present a literature review that examines agroecology to contribute to climate change adaptation and resource depletion in the Mediterranean region.	Agroecology definition.	This review shows that agroecology can provide a systemic approach to offer a holistic framework that allows the recovery and assessment of traditional knowledge and the creation of new local knowledge to enhance resilience.

Table 2. *Cont.*

Reference	Objective	Definitions, Aspects, and Characteristics of the Concepts of Agroecology and Sustainable Agriculture	Results
Ewert et al. [55]	To explore how agroecology can become the primary model for transforming agriculture into more sustainable and resilient agri-food systems within the current economic and political context.	Agroecology definition.	Improving agroecology requires a comprehensive systems approach that addresses individual farms at regional and global levels. This approach should consider processes, relationships, stakeholders, drivers of change, sustainability indicators, and assessment methods at each scale.
Mockshell and Kamanda [45]	To examine the debate on intensified agroecology and intensified sustainable agriculture.	Definition of agroecology and sustainable agriculture. Identification of actors, vision, characteristics, and objectives.	The findings indicate notable differences in the composition of actors involved in the Sustainable Agricultural Intensification (SAI) and Agricultural Ecosystem Innovation (AEI) pathways. Both pathways are designed to enhance food security through the optimal and sustainable utilization of resources.
Muhie [56]	To test various approaches and practices to demonstrate that their adoption can lead to agricultural sustainability.	Sustainable agriculture definition.	Results show that adopting various approaches and practices can safeguard agricultural sustainability.
Trigo, Marta-Costa and Fragoso [57]	To provide a reference framework for decision-making regarding sustainable agriculture based on identifying four principles: integrated management, dynamic balance, regenerative design, and social development.	Sustainable agriculture definition.	A methodology to assess a specific behavior, process, or situation and implement practices derived from sustainable agriculture.
Ogemah [58]	To address divergent perceptions of sustainable agriculture that hinder the universal application of the concept in the agricultural sector and could negatively impact the achievement of sustainable development goals.	Definition of sustainable agriculture and its principles.	The author proposed a common understanding of the concept of sustainable agriculture and contributed to the modernization process of agriculture in Africa.
Amrit, Raghavendra and Minakshi [61]	To analyze the definition of sustainable agriculture and how it helps address the impacts of climate change.	Definition of sustainable agriculture and its role in combating climate change.	The authors identified India's prevailing challenges and opportunities based on the sustainable and climate-smart agriculture analysis.
Gil et al., 2022 [60]	To present a good viticulture practice at an advanced level of agroecological transition.	Definition of agroecology and its elements.	Results express improvements in this agroecosystem's energy efficiency and financial returns compared to conventional farms in the region.

6. Discussion

The literature review showed that the authors' analysis of agroecology and sustainable agriculture concepts was quite general. Therefore, it is essential to explore further how they defined these concepts, the disciplines involved, their perspectives, and the key players, among other aspects. Ogemah [58] and Mockshell and Kamanda [45] have tried to clarify the definitions of both concepts and identify their components.

Although the reviewed works agree that both concepts respond to more than one discipline and must maintain a systemic [45,54,57,58] and ethical approach, according to Caporali [12], they are, particularly those that define agroecology, because the importance of a holistic vision is included [54], employing multiple disciplines for rural development and considering all environmental and human elements [45].

A second component identified in the definitions of these concepts is that of the actors, which, in the case of agroecology, are non-governmental organizations (NGOs), civil society, researchers, and academics [45]. Meanwhile, sustainable agriculture integrates traditional and modern scientific knowledge, according to Amrit, Minakshi, and Raghavendra [61]. This integration includes continuous coordination among various actors, disciplines, and elements to enhance the environmental, social, and economic aspects of the agri-food system without neglecting the participation of government agencies [45].

In another sense, Aguilera et al. [54] highlight a determining factor in the definition of agroecology, the scale. Authors emphasize that agroecology takes a holistic approach, considering the agricultural scale and the entire food system, the literature indicates that agroecological practices are expanding and being replicated in other rural production units, contributing to the sustainability of farming systems. However, it is unclear whether agroecology practices and strategies can be applied to a broader system on a larger scale, involving more elements and participants.

The statement is also supported by Mockshell and Kamanda [45] and Gil et al. [60], who discuss the issue of scale or scalability in their research. They mention that this aspect of agroecology involves implementing and adjusting high-performance technologies to meet the increasing global food demand while minimizing environmental impacts. However, they have not yet specified whether agroecological practices can be applied to diverse agricultural systems of various compositions and on a large scale.

It should be noted that the scale is a fundamental element in developing countries, according to Mockshell and Kamanda [45], whose farmers lack access to external inputs and often face food insecurity. They emphasize that agroecology can address the production challenges faced by small farmers and meet the dietary needs of communities in these regions. This aligns with the findings of Altieri [74], who stresses that agroecology is crucial in ensuring food security in rural areas.

However, Ewert et al. [55] consider that adopting agroecological practices could result in an economically uncertain transition for small producers, which hinders their widespread adoption by other small producers. The above is because, according to their findings, relatively few studies have tried to empirically show the impact of agroecological management on the economic performance of farms, considering all costs, benefits, and externalities.

Regarding sustainable agriculture, Ogemah [58] highlights that it focuses on improving productivity to ensure food security without significantly impacting other sustainable development goals. To this end, he offers an interesting review in which he argues what this concept is and is not. Some other authors consider additional elements essential for defining sustainable agriculture, such as ecology and technology [56]. In a broad sense, the latter involves traditional knowledge, as mentioned above, but also the use of GMOs [45].

The reviewed articles discuss agroecology and sustainable agriculture, but it is important to note that there is no consensus on the precise definition of these concepts. Additionally, some authors have questioned the additional disciplines incorporated into agroecology over the years, as the term was initially derived only from agronomy and ecology. Similarly, no studies specifically identify the disciplines that were initially part of sustainable agriculture.

Regarding the involvement of actors in agroecology, the inclusion of decision-makers is not explicit. While the improvement in production through agroecological practices is discussed, in terms of food security and sovereignty, general planning is necessary to ensure the fulfillment of specific regional and/or national goals for producers and consumers.

One aspect that remains to be defined in agroecology is how it could specifically address scalability, which is most evident in that it can be managed through sustainable agriculture. However, concerning this last concept, the analyzed definitions offer few

alternatives to using technologies that promote achieving goals such as those established in the SDGs.

This article is consistent with Ogemah [58], Hrabanski and Le Coq [59], and Mockshell and Kamanda [45], who provide a more detailed examination of agroecology and sustainable agriculture, respectively. They particularly question the intended purpose of each concept. However, these proposals emphasize the importance of promoting global food security.

While research mentions scale and scope aspects, few studies specifically highlight how these differences impact the practical implementation and efficiency of agri-food systems. This literature review can support an appropriate choice between the concepts of agroecology or sustainable agriculture, depending on the needs and characteristics of the context in a decision-making framework on agricultural policies, the design of rural development programs, or strategies for adaptation to climate change.

Although this review aims to explore the concepts of agroecology and sustainable agriculture by identifying the contributions of various authors, these concepts are shaped through the construction of knowledge and enriched by discoveries. A limitation in the review process was that not all potentially eligible studies could be accessed due to them not being open access. The review was also restricted to English and Spanish, which may have excluded valuable contributions in other languages. The commentary aims to clarify the presentation of results and facilitate a better discussion of the findings.

7. Conclusions

The concept of agroecology is recognized as a fundamental approach to steering agriculture towards sustainability and resilience. It seeks to maintain a holistic perspective that encompasses the ecological, economic, and social aspects of the food system, as well as addressing the issue of food insecurity, but primarily favors the environmental aspect of sustainable development.

For agroecology to achieve the required impact, it is crucial to adopt a multi-scale systems approach that integrates processes, stakeholders, sustainability indicators, and evaluation methods at all levels. These strategies are closely related primarily to sustainable agriculture, which is why agroecology is just a tiny part of the broader concept of sustainable agriculture.

Sustainable agriculture ensures food security by promoting environmental conservation and efficient food production. It is not that agroecology is irrelevant to this vital task; instead, sustainable agriculture is better equipped to address persistent challenges such as the impact of climate change, inadequate policies, and unsustainable practices due to its diverse actors, elements, and strategies.

This approach is necessary because it involves many variables and indicators that align with the paradigms of complexity and structuralism. Consequently, a technique suitable for this mixed perspective is required. Based on experience, this research work applied the MaxQDA program for practical analysis. These methodological procedures then allow the generation of information to promptly address agroecology, the agroecological transition, and the achievement of sustainable agriculture.

Future research should include critical reviews of agroecology and sustainable agriculture concepts. Additionally, scholars must specify the stance taken by each research group and the reasons behind it. Researchers in these fields should consider that agroecology is limited to agronomy and ecology. In contrast, with its systemic and integral approach, sustainable agriculture encompasses various disciplines, actors, strategies, and, very importantly, scalability.

Finally, although this work is not aimed at proposing policies derived from the conclusions, it is significant to mention that in a context such as Mexico, agroecology must be present in the definition of agricultural policies since a large number of producers are small and medium-sized; however, agroecology presents limitations to advancing towards sustainable agriculture. Therefore, the agroecological transition for agricultural sustainability requires a policy that privileges those agroecological practices consistent with scale, sovereignty, and food security so that agricultural systems advance toward sustainability.

Supplementary Materials: The following supporting information can be downloaded at: https://www.mdpi.com/article/10.3390/su17051805/s1, Table S1: The PRISMA checklist. Reference [75] is cited in the Supplementary Materials.

Author Contributions: Conceptualization, K.T.-S. and J.M.R.-P.; methodology K.T.-S., M.C.G.-L. and J.M.R.-P.; validation, J.M.R.-P.; formal analysis, J.M.R.-P., I.V.-A., M.Á.M.-T. and D.F.-G.; investigation, K.T.-S., M.C.G.-L., M.L.M.-d.l.P., D.F.-G., J.A.G.-F. and P.C.E.-L.; writing—original draft preparation, M.L.M.-d.l.P., D.F.-G., J.A.G.-F. and P.C.E.-L.; writing—review and editing, K.T.-S. and J.C.T.-A.; visualization J.M.R.-P., I.V.-A., M.Á.M.-T. and J.C.T.-A.; supervision, J.M.R.-P.; project acquisition, I.V.-A. and M.Á.M.-T. All authors have read and agreed to the published version of the manuscript.

Funding: This research was funded by Project 322596 of the National Council of Humanities, Science, and Technology of Mexico (CONAHCYT), along with postdoctoral fellowships and support from the Research Center for Food and Development (CIAD).

Conflicts of Interest: The authors declare no conflicts of interest.

References

1. Moraga Quezada, M.E. Enmiendas Orgánicas y Sintéticas y su Efecto en la Producción de Maíz (*Zea mays* L.) y Frijol (*Phaseolus vulgaris* L.) y en la Fertilidad del Suelo. Ph.D. Thesis, Universidad Nacional Agraria, Lima, Peru, 2021.
2. Bensin, B.M. Agroecological Characteristics Description and Classification of the Local Corn Varieties-Chorotypes. Prague, 1925.
3. Klages, K.H.W. Crop Ecology and Ecological Crop Geography in the Agronomic Curriculum. *Agron. J.* **1928**, *20*, 336–353. [CrossRef]
4. Gallardo-López, F.; Hernández-Chontal, M.A.; Cisneros-Saguilán, P.; Linares-Gabriel, A. Development of the Concept of Agroecology in Europe: A Review. *Sustainability* **2018**, *10*, 1210. [CrossRef]
5. Wezel, A.; Herren, B.G.; Kerr, R.B.; Barrios, E.; Gonçalves, A.L.R.; Sinclair, F. Agroecological Principles and Elements and Their Implications for Transitioning to Sustainable Food Systems. A Review. *Agron. Sustain. Dev.* **2020**, *40*, 1–13. [CrossRef]
6. Mouratiadou, I.; Wezel, A.; Kamilia, K.; Marchetti, A.; Paracchini, M.L.; Bàrberi, P. The Socio-Economic Performance of Agroecology. A Review. *Agron. Sustain. Dev.* **2024**, *4*, 19. [CrossRef]
7. Janzen, D.H. Tropical Agroecosystems. *Science* **1973**, *182*, 1212–1219. [CrossRef] [PubMed]
8. Gliessman, S.R.; Garcia-Espinosa, R.; Amador, M. The Ecological Basis for the Application of Traditional Agricultural Technology in the Management of Tropical Agro-Ecosystems. *Agro-Ecosyst.* **1981**, *7*, 173–185. [CrossRef]
9. Levidow, L.; Sansolo, D.; Schiavinatto, M. Agroecological Innovation Constructing Socionatural Order for Social Transformation: Two Case Studies in Brazil. *Tapuya Lat. Am. Sci. Technol. Soc.* **2021**, *4*, 288–307. [CrossRef]
10. Ruiz-Rosado, O. Agroecología: Una Disciplina Que Tiende a La Transdisciplina. *Interciencia* **2006**, *31*, 140–145.
11. Casanova, L.; Martínez, J.; López, S.; López, G. De von Bertalanffy a Luhmann: Deconstrucción del Concepto "Agroecosistema" a Través de las Generaciones Sistémicas. *Rev. Mad.* **2016**, *35*, 60–74. [CrossRef]
12. Caporali, F. Agroecology as a Science of Integration for Sustainability in Agriculture. *Ital. J. Agron.* **2011**, *2*, 73–82. [CrossRef]
13. Fernández, L. Caracterización de la Comercialización de Bolsones Agroecológicos: Estudio de Caso En La Plata, 2019–2020. *Rev. Huellas* **2021**, *25*, 193–209. [CrossRef]
14. Altieri, M.A. *Agroecology: The Science of Sustainable Agriculture*, 2nd ed.; Westview Press: Boulder, CO, USA, 1995.
15. Gliessman, S.R. *Agroecology: Researching the Ecological Basis for Sustainable Agriculture*; Springer: New York, NY, USA, 1990.
16. Dalgaard, T.; Hutchings, N.J.; Porter, J.R. Agroecology, Scaling and Interdisciplinarity. *Agric. Ecosyst. Environ.* **2003**, *100*, 39–51. [CrossRef]
17. Francis, C.; Lieblein, G.; Gliessman, S.; Breland, T.A.; Creamer, N.; Harwood, R.; Salomonsson, L.; Helenius, J.; Rickerl, D.; Salvador, R.; et al. Agroecology: The Ecology of Food Systems. *J. Sustain. Agric.* **2003**, *22*, 99–118. [CrossRef]
18. Gliessman, S.R. Transforming Food Systems with Agroecology. *Agroecol. Sustain. Food Syst.* **2016**, *40*, 187–189. [CrossRef]

19. Wezel, A.; Bellon, S.; Doré, T.; Francis, C.; Vallod, D.; David, C. Agroecology as a science, a movement and a practice. A review. *Agron. Sustain. Dev.* **2009**, *29*, 503–515. [CrossRef]
20. Bezner, R.; Madsen, S.; Stüber, M.; Liebert, J.; Enloe, S.; Borghino, N.; Parros, P.; Mutyambai, D.M.; Prudhon, M.; Wezel, A. Can Agroecology Improve Food Security and Nutrition? A Review. *Glob. Food Secur.* **2021**, *29*, 100540. [CrossRef]
21. Méndez, V.E.; Bacon, C.M.; Cohen, R. La Agroecología como un Enfoque Transdisciplinar, Participativo y Orientado a la Acción. *Agroecología* **2013**, *8*, 9–18. Available online: https://revistas.um.es/agroecologia/article/view/212061 (accessed on 15 August 2024).
22. Galt, R.E.; Pinzón, N.; Robinson, N.I.; Baukloh, M.B. Agroecology and the Social Sciences: A Half-Century Systematic Review. *Agric. Syst.* **2024**, *216*, 103881. [CrossRef]
23. Caporali, F. Agroecology as a Transdisciplinary Science for a Sustainable Agriculture. In *Biodiversity, Biofuels, Agroforestry, and Conservation Agriculture. Sustainable Agriculture Reviews*; Lichtfouse, E., Ed.; Springer: Dordrecht, The Netherlands, 2011; Volume 5, pp. 1–71. ISBN 978-90-481-9512-1.
24. Laforge, J.M.; Dale, B.; Levkoe, C.Z.; Ahmed, F. The Future of Agroecology in Canada: Embracing the Politics of Food Sovereignty. *J. Rural Stud.* **2021**, *81*, 194–202. [CrossRef]
25. López-García, D.; Benlloch, L.; Calabuig, V.; Carucci, P.; Diez, I.; Herrero, A.; López, M.; Pérez, J.M.; Vicente-Amazán, L. Las Transiciones Hacia la Sostenibilidad como Procesos de Final Abierto: Dinamización Local Agroecológica con Horticultores Convencionales de l'Horta de València. In *Boletín de la Asociación Española de Geografía*; Asociación Española de Geografía location: Madrid, Spain, 2021; Volume 88. [CrossRef]
26. Anderson, C.; Anderson, M. Looking Forward: Resources to Inspire a Transformational Agroecology. In *Transformation of Our Food System: The Making of a New Paradigm*; Herren, H.R., Haerlin, B., IAASTD+10 Advisory Group, Eds.; Foundation on Future Farming and Biovision: Zurich, Switzerland, 2020; pp. 169–180.
27. Galt, R.E.; Parr, D.; Van Soelen Kim, J.; Beckett, J.; Lickter, M.; Ballard, H. Transformative Food Systems Education in a Land-Grant College of Agriculture: The Importance of Learner-Centered Inquiries. *Agric. Hum. Values* **2013**, *30*, 129–142. [CrossRef]
28. Méndez, V.E.; Bacon, C.M.; Cohen, R.; Gliessman, S.R. *Agroecology: A Transdisciplinary, Participatory, and Action-Oriented Approach*, 1st ed.; CRC Press: Boca Raton, FL, USA; Tylor & Francis Group: Abingdon, UK, 2016; ISBN 9780367436018.
29. Rosset, P.M.; Altieri, M.A. *Agroecology: Science and Politics*; Fernwood Publishing: Newberg, OR, USA, 2017; ISBN 978-185339-994-4.
30. Rossi, G. Exploring Agroecological Sustainability: Unearthing Innovators and Documenting a Community of Practice in Southeast Australia. *Soc. Nat. Resour.* **2020**, *30*, 585–600. [CrossRef]
31. Bensin, B.M. Possibilities for International Cooperation in Agroecological Investigations. *Int. Rev. Agr.* **1930**, *21*, 277–284.
32. Klages, K.H.W. *Ecological Crop Geography*; Macmillan Company: New York, NY, USA, 1942; p. 615. [CrossRef]
33. Altieri, M.A. Agroecology: A New Research and Development Paradigm for World Agriculture. *Agric. Ecosyst. Environ.* **1989**, *27*, 37–46. [CrossRef]
34. Raigón, M.D. El Valor de la Agroecología para Promover Sistemas Alimentarios Urbanos Sostenibles. *Kult-Ur* **2023**, *10*, 127–153. [CrossRef]
35. Dönmez, D.; Isak, M.A.; İzgü, T.; Şimşek, Ö. Green Horizons: Navigating the Future of Agriculture through Sustainable Practices. *Sustainability* **2024**, *16*, 3505. [CrossRef]
36. Terán, K. Equidad Gerencial, Como una Demanda Intangible de Mercado: Hacia Organizaciones Sustentables, el Caso de Uva de Mesa. Ph.D. Thesis, Centro de Investigación en Alimentación y Desarrollo, A.C., Hermosillo, Sonora, Mexico, 2022. Available online: https://ciad.repositorioinstitucional.mx/jspui/bitstream/1006/1105/1/Karla%20Ter%C3%A1n%20Samaniego.pdf (accessed on 15 August 2024).
37. López, I.L. Sobre el Desarrollo Sostenible y la Sostenibilidad: Conceptualización y Crítica. Barataria. *Rev. Castell. Manchega Cienc. Soc.* **2015**, *20*, 111–128. [CrossRef]
38. Rizo-Mustelier, M.; Vuelta-Lorenzo, D.R.; Lorenzo-García, A.M. Agricultura, Desarrollo Sostenible, Medioambiente, Saber Campesino y Universidad. *Cienc. PC* **2017**, *2*, 106–120. Available online: https://www.redalyc.org/journal/1813/181351615008/html/ (accessed on 15 August 2024).
39. Martínez, D.G. Agroecología, Organizaciones y Afectos. Las Intervenciones de Técnicos Agrónomos en el Cinturón Hortícola Platense (Argentina). *Eutopía Rev. Desarro. Económico Territ.* **2022**, *22*, 125–141. [CrossRef]
40. Contreras, A.C. Certificaciones Agrícolas como Conducta Estratégica y el Alcance de Mercado de las Firmas que Forman el Sistema de Vid de Mesa Sonorense. Ph.D. Thesis, Centro de Investigación en Alimentación y Desarrollo, A.C., Hermosillo, Sonora, Mexico, 2019. Available online: https://ciad.repositorioinstitucional.mx/jspui/bitstream/1006/1099/3/2020_Ana%20Carolina%20Contreras%20Valenzuela.pdf (accessed on 15 August 2024).
41. FAO. Scaling Up Agroecology to Achieve the SDGs; Centro de Conocimiento sobre Agroecología, Organización de las Naciones Unidas para la Agricultura y la Alimentación. 2024. Available online: https://www.fao.org/agroecology/overview/agroecology-and-the-sustainable-development-goals/es/ (accessed on 15 August 2024).

42. Janker, J.; Mann, S.; Rist, S. What Is Sustainable Agriculture? Critical Analysis of the International Political Discourse. *Sustainability* **2018**, *10*, 4707. [CrossRef]
43. MacDonald, R.; Reitmeier, C. *Understanding Food Systems: Agriculture, Food Science, and Nutrition in the United States*; Academic Press: Cambridge, MA, USA, 2017; ISBN 9780128044452. eBook ISBN 9780128044858.
44. González-León, Y.; Anducho-Reyes, M.A.; Cartagena-Luna, A.; Mercado-Flores, Y. Sustainable Agriculture: A Tool for Food Sovereignty. In *Handbook T-VIII*; ECORFAN-México S.C.: Ciudad de México, Mexico, 2020; Chapter 10; ISBN 978-607-8695-32-4.
45. Mockshell, J.; Kamanda, J. Beyond the Agroecological and Sustainable Agricultural Intensification Debate: Is Blended Sustainability the Way Forward? *Int. J. Agric. Sustain.* **2018**, *16*, 127–149. [CrossRef]
46. Quispe-Ojeda, T.C. La Agroecología como Alternativa para el Desarrollo Sostenible y Sustentable. *Cienciamatria* **2022**, *8*, 33–45. [CrossRef]
47. CEPAL. La Digitalización al Servicio de la Transición Agroecológica; Comisión Económica para América Latina y el Caribe. 2019. Available online: https://www.cepal.org/sites/default/files/news/files/23-00546-folleto-la_digitalizacion_web_0.pdf (accessed on 15 August 2024).
48. Do Nascimento, F.S.; Calle-Collado, Á.; Benito, R.M. Economía Social y Solidaria y Agroecología en Cooperativas de Agricultura Familiar en Brasil como Forma de Desarrollo de una Agricultura Sostenible. *Ciriec-España* **2020**, *98*, 189–211. [CrossRef]
49. Barrios Latorre, S.A.; Sadovska, V.; Chongtham, I.R. Perspectives on Agroecological Transition: The Case of Guachetá Municipality, Colombia. *Agroecol. Sustain. Food Syst.* **2023**, *47*, 382–412. [CrossRef]
50. Bisht, I.S.; Rana, J.C.; Jones, S.; Estrada-Carmona, N.; Yadav, R. Agroecological Approach to Farming for Sustainable Development: The Indian Scenario. In *Biodiversity of Ecosystems*; IntechOpen: London, UK, 2022; pp. 107–135. [CrossRef]
51. Dreifuss-Serrano, C.; Schreier-Barreto, C.; Jumpa-Zevallos, M. *Criterios de Clasificación de la Vivienda Informal: Una Revisión Sistemática PRISMA como Herramienta para Establecimiento y Análisis de Categorías*; Universidad de Lima: Lima, Peru, 2018; Volume 10, pp. 98–103.
52. Moher, D.; Liberati, A.; Tetzlaff, J.; Altman, D.G. Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. *Int. J. Surg.* **2010**, *8*, 336–341. [CrossRef] [PubMed]
53. Sánchez-Teruel, D.; Robles-Bello, M.A. Instrumentos de Evaluación en Inteligencia Emocional: Una Revisión Sistemática Cuantitativa. *Perspect. Educ.* **2018**, *57*, 27–50. [CrossRef]
54. Aguilera, E.; Díaz-Gaona, C.; García-Laureano, R.; Reyes-Palomo, C.; Guzmán, G.I.; Ortolani, L.; Sánchez-Rodríguez, M.; Rodríguez-Estévez, F.V. Agroecology for Adaptation to Climate Change and Resource Depletion in the Mediterranean Region: A Review. *Agric. Syst.* **2020**, *181*, 102809. [CrossRef]
55. Ewert, F.; Baatz, R.; Finger, R. Agroecology for a Sustainable Agriculture and Food System: From Local Solutions to Large-Scale Adoption. *Annu. Rev. Resour. Econ.* **2023**, *15*, 351–381. [CrossRef]
56. Muhie, S.H. Novel Approaches and Practices to Sustainable Agriculture. *J. Agric. Food Res.* **2022**, *10*, 100446. [CrossRef]
57. Trigo, A.; Marta-Costa, A.; Fragoso, R. Principles of Sustainable Agriculture: Defining Standardized Reference Points. *Sustainability* **2021**, *13*, 4086. [CrossRef]
58. Ogemah, V.K. Sustainable Agriculture: Developing a Common Understanding for Modernization of Agriculture in Africa. *Afr. J. Food Agric. Nutr. Dev.* **2017**, *17*, 11673–11690. [CrossRef]
59. Hrabanski, M.; Le Coq, J.F. Climatisation of Agricultural Issues in the International Agenda through Three Competing Epistemic Communities: Climate-Smart Agriculture, Agroecology, and Nature-Based Solutions. *Environ. Sci. Policy* **2022**, *127*, 311–320. [CrossRef]
60. Gil, J.; Alter, E.; La Rota, M.J.; Tello, E.; Galletto, V.; Padró, R.; Martínez, T.; Darnay, S.; Marull, J. Towards an Agroecological Transition in the Mediterranean: A Bioeconomic Assessment of Viticulture Farming. *J. Clean. Prod.* **2022**, *380*, 134999. [CrossRef]
61. Amrit, M.; Minakshi, K.; Raghavendra, K.J. Sustainable and Climate Smart Agriculture: Challenges and Opportunities in Indian Perspective. *Agriallis E-Newsl.* **2021**, *3*, 47–57.
62. Fernández, C.; Ollivier, G.; Bellon, S. Transdisciplinarity in Agroecology: Practices and Perspectives in Europe. *Agroecol. Sustain. Food Syst.* **2020**, *45*, 523–550. [CrossRef]
63. Isaac, M.E.; Sinclair, F.; Laroche, G.; Olivier, A.; Thapa, A. The Ties that Bind: How Trees Can Enhance Agroecological Transitions. *Agroforest. Syst.* **2024**, *98*, 2369–2383. [CrossRef]
64. Fonseca, A.F.; Polita, F.; Madureira, L. How Agroecological Transition Frameworks Are Reshaping Agroecology: A Review. *Land* **2024**, *13*, 1930. [CrossRef]
65. Padró, R.; Tello, E. Exploring Agroecology Transition Scenarios: A Pfaundler's Spectrum Assessment on the Relocation of Agri-Food Flows. *Land* **2022**, *11*, 824. [CrossRef]
66. Anderson, M.D.; Rivera-Ferre, M. Food system narratives to end hunger: Extractive versus regenerative. *Curr. Opin. Environ. Sustain.* **2021**, *49*, 18–25. [CrossRef]
67. Nunes da Silva, M.; Machado, J.; Osorio, J.; Duarte, R.; Santos, C.S. Non-Essential Elements and Their Role in Sustainable Agriculture. *Agronomy* **2022**, *12*, 888. [CrossRef]

68. Riccaboni, A.; Neri, E.; Trovarelli, F.; Pulselli, R.M. Sustainability-oriented research and innovation in 'farm to fork' value chains. *Curr. Opin. Food Sci.* **2021**, *42*, 102–112. [CrossRef]
69. Boros, A.; Szólik, E.; Desalegn, G.; Tőzsér, D. A Systematic Review of Opportunities and Limitations of Innovative Practices in Sustainable Agriculture. *Agronomy* **2025**, *15*, 76. [CrossRef]
70. Tiwari, P.; Park, K.-I. Seed Biotechnologies in Practicing Sustainable Agriculture: Insights and Achievements in the Decade 2014–2024. *Appl. Sci.* **2024**, *14*, 11620. [CrossRef]
71. Brunelle, T.; Chakir, R.; Carpentier, A.; Dorin, B.; Goll, D.; Guilpart, N.; Tang, F. Reducing Chemical Inputs in Agriculture Requires a System Change. *Commun. Earth Environ.* **2024**, *5*, 369. [CrossRef]
72. Candel, J.; Pe'er, G.; Finger, R. Science Calls for Ambitious European Pesticide Policies. *Nat. Food* **2023**, *4*, 272. [CrossRef] [PubMed]
73. Pisante, M.; Gogoi, N.; Farooq, M. (Eds.) Sustainability, sustainable agriculture, and the environment. In *Sustainable Agriculture and the Environment*; Academic Press: Cambridge, MA, USA, 2023; pp. 3–24. [CrossRef]
74. Altieri, M.A. Agroecology: The Science of Natural Resource Management for Poor Farmers in Marginal Environments. *Agric. Ecosyst. Environ.* **2002**, *93*, 1–24. [CrossRef]
75. Page, M.J.; McKenzie, J.E.; Bossuyt, P.M.; Boutron, I.; Hoffmann, T.C.; Mulrow, C.D.; Shamseer, L.; Tetzlaff, J.M.; Akl, E.A.; Brennan, S.E.; et al. The PRISMA 2020 statement: An updated guideline for reporting systematic reviews. *BMJ* **2021**, *372*, n71. [CrossRef] [PubMed]

Disclaimer/Publisher's Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.